

Lecture - 9

CO4: Compare various statistical methods of study data samples

CO5: Analyze and evaluation of different sets of data using hypothesis testing.

REGRESSION

Meaning

A study of measuring the relationship between associated variables, wherein one variable is dependent on another independent variable, called as Regression. It is developed by Sir Francis Galton in 1877 to measure the relationship of height between parents and their children.

Regression analysis is a statistical tool to study the nature and extent of functional relationship between two or more variables and to estimate (or predict) the unknown values of dependent variable from the known values of independent variable.

The variable that forms the basis for predicting another variable is known as the Independent Variable and the variable that is predicted is known as dependent variable. For example, if we know that two variables price (X) and demand (Y) are closely related we can find out the most probable value of X for a given value of Y or the most probable value of Y for a given value of X.

Similarly, if we know that the amount of tax and the rise in the price of a commodity are closely related, we can find out the expected price for a certain amount of tax levy.

Uses of Regression Analysis:

1. It provides estimates of values of the dependent variables from values of independent variables.
2. It is used to obtain a measure of the error involved in using the regression line as a basis for estimation.
3. With the help of regression analysis, we can obtain a measure of degree of association or correlation that exists between the two variables.
4. It is highly valuable tool in economies and business research, since most of the problems of the economic analysis are based on cause and effect relationship.

Distinction between Correlation and Regression

S. No.	Correlation	Regression
1	It measures the degree and direction of relationship between the variables.	It measures the nature and extent of average relationship between two or more variables in terms of the original units of the data
2	It is a relative measure showing association between the variables.	It is an absolute measure of relationship.
3	Correlation Coefficient is independent of change of both origin and scale.	Regression Coefficient is independent of change of origin but not scale.
4	Correlation Coefficient is independent of units of measurement.	Regression Coefficient is not independent of units of measurement.

5	Expression of the relationship between the variables ranges from -1 to $+1$	Expression of the relationship between the variables may be in any of the forms like: $Y=a+bX$ $Y=a+bX+cX^2$
6	It is not a forecasting device.	It is a forecasting device which can be used to predict the value of dependent variable from the given value of independent variable.
7	There may be zero correlation such as weight of wife and income of husband.	There is nothing like zero regression.

Regression Lines and Regression Equation

Regression lines and regression equations are used synonymously. Regression equations are algebraic expression of the regression lines. Let us consider two variables: X & Y . If y depends on x , then the result comes in the form of simple regression. If we take the case of two variable X and Y , we shall have two regression lines as the regression line of X on Y and regression line of Y on X . The regression line of Y on X gives the most probable value of Y for given value of X and the regression line of X on Y given the most probable value of X for given value of Y . Thus, we have two regression lines. However, when there is either perfect positive or perfect negative correlation between the two variables, the two regression line will coincide, i.e. we will have one line. If the variables are independent, r is zero and the lines of regression are at right angles i.e. parallel to X axis and Y axis.

Therefore, with the help of simple linear regression model we have the following two regression lines

1. Regression line of Y on X: This line gives the probable value of Y (Dependent variable) for any given value of X (Independent variable).

$$\text{Regression line of Y on X} \quad : \quad Y - \bar{Y} = b_{YX}(X - \bar{X})$$

$$\text{Or} \quad : \quad Y = a + bX$$

2. Regression line of X on Y: This line gives the probable value of X (Dependent variable) for any given value of Y (Independent variable).

$$\text{Regression line of X on Y} \quad : \quad X - \bar{X} = b_{XY}(Y - \bar{Y})$$

$$\text{Or} \quad : \quad X = a + bY$$

In the above two regression lines or regression equations, there are two regression parameters, which are “a” and “b”. Here “a” is unknown constant and “b” which is also denoted as “ b_{yx} ” or “ b_{xy} ”, is also another unknown constant popularly called as regression coefficient. Hence, these “a” and “b” are two unknown constants (fixed numerical values) which determine the position of the line completely. If the value of either or both of them is changed, another line is determined. The parameter “a” determines the level of the fitted line (i.e., the distance of the line directly above or below the origin). The parameter “b” determines the slope of the line (i.e., the change in Y for unit change in X).

If the values of constants “a” and “b” are obtained, the line is completely determined. But the question is how to obtain these values. The answer is provided by the method of least squares. With the little algebra and differential

calculus, it can be shown that the following two **normal equations**, if solved simultaneously, will yield the values of the parameters “a” and “b”.

Regression Coefficient:

The quantity “b” in the regression equation is called as the regression coefficient or slope coefficient. Since there are two regression equations, therefore, we have two regression coefficients.

1. Regression Coefficient X on Y, symbolically written as “ b_{xy} ”.
2. Regression Coefficient Y on X, symbolically written as “ b_{yx} ”.

Different formula's used to compute regression coefficients:

Method	Regression Coefficient X on Y	Regression Coefficient Y on X
Using the correlation coefficient (r) and standard deviation (σ)	$b_{xy} = r \frac{\sigma_x}{\sigma_y}$	$b_{yx} = r \frac{\sigma_y}{\sigma_x}$
Direct Method: Using sum of X and Y	$b_{xy} = \frac{N\sum XY - \sum X \sum Y}{N\sum Y^2 - (\sum Y)^2}$	$b_{yx} = \frac{N\sum XY - \sum X \sum Y}{N\sum X^2 - (\sum X)^2}$
When deviations are taken from arithmetic mean	$b_{xy} = \frac{\sum xy}{\sum y^2}$ Where $x = X - \bar{X}$	$b_{yx} = \frac{\sum xy}{\sum x^2}$ Where $y = Y - \bar{Y}$

Properties of Regression Coefficients

1. The coefficient of correlation is the geometric mean of the two regression coefficients. Symbolically $r = \sqrt{b_{xy} * b_{yx}}$
2. If one of the regression coefficients is greater than unity, the other must be less than unity, since the value of the coefficient of correlation cannot exceed unity. For example, if $b_{xy} = 1.2$ and $b_{yx} = 1.4$ “r” would be $= \sqrt{1.2 * 1.4} = 1.29$, which is not possible.
3. Both the regression coefficient will have the same sign. i.e., they will be either positive or negative. In other words, it is not possible that one of the regression coefficients is having minus sign and the other plus sign.
4. The coefficient of correlation will have the same sign as that of regression coefficient, i.e., if regression coefficient has a negative sign, “r” will also have negative sign and if the regression coefficient has a positive sign, “r” would also be positive. For example, if $b_{xy} = -0.2$ and $b_{yx} = -0.8$ then $r = -\sqrt{0.2 * 0.8} = -0.4$
5. The average value of the two-regression coefficient would be greater than the value of coefficient of correlation. In symbol $(b_{xy} + b_{yx})/2 > r$. For example, if $b_{xy} = 0.8$ and $b_{yx} = 0.4$ then average of the two values $= (0.8 + 0.4) / 2 = 0.6$ and the value of $r = \sqrt{0.8 * 0.4} = 0.566$ which less than 0.6.
6. Regression coefficients are independent of change of origin but not scale.

Example 1: Find the two-regression equation of X on Y and Y on X from the following data:

X	10	12	16	11	15	14	20	22
Y	15	18	23	14	20	17	25	28

Solution: Calculation of Regression Equation

X	Y	x^2	y^2	XY
10	15	100	225	150
12	18	144	324	216
16	23	256	529	368
11	14	121	196	154
15	20	225	400	300
14	17	196	289	238
20	25	400	625	500
22	28	484	784	616
120	160	1,926	3,372	2,542
ΣX	ΣY	ΣX^2	ΣY^2	ΣXY

Here N = Number of elements in either series X or series Y = 8

Now we will proceed to compute regression equations using normal equations.

Regression equation of X on Y: $X = a + bY$

The two normal equations are:

$$\Sigma X = Na + b\Sigma Y$$

$$\Sigma XY = a\Sigma Y + b\Sigma Y^2$$

Substituting the values in above normal equations, we get

$$120 = 8a + 160b \quad \dots (i)$$

$$2542 = 160a + 3372b \quad \dots (ii)$$

Let us solve these equations (i) and (ii) by simultaneous equation method
 Multiply equation (i) by 20 we get $2400 = 160a + 3200b$

Now rewriting these equations:

$$\begin{array}{r} 2400 = 160a + 3200b \\ 2542 = 160a + 3372b \\ (-) \quad (-) \quad (-) \quad \cdot \\ \hline -142 = \quad \quad -172b \end{array}$$

Therefore, now we have $-142 = -172b$, this can rewrite as $172b = 142$

Now, $b = 142/172 = 0.8256$ (rounded off)

Substituting the value of b in equation (i), we get

$$\begin{array}{rcl} 120 & = & 8a + (160 * 0.8256) \\ 120 & = & 8a + 132 \text{ (rounded off)} \\ 8a & = & 120 - 132 \\ 8a & = & -12 \\ a & = & -12/8 \\ a & = & -1.5 \end{array}$$

Thus we got the values of $a = -1.5$ and $b = 0.8256$

Hence the required regression equation of X on Y :

$$X = a + bY \Rightarrow X = -1.5 + 0.8256Y$$

Regression equation of Y on X : $Y = a + bX$

The two normal equations are:

$$\begin{array}{rcl} \sum Y & = & Na + b\sum X \\ \sum XY & = & a\sum X + b\sum X^2 \end{array}$$

Substituting the values in above normal equations, we get

$$160 = 8a + 120b \quad \dots (iii)$$

$$2542 = 120a + 1926b \quad \dots (iv)$$

Let us solve these equations (iii) and (iv) by simultaneous equation method
 Multiply equation (iii) by 15 we get $2400 = 120a + 1800b$

Now rewriting these equations:

$$\begin{array}{r} 2400 = 120a + 1800b \\ 2542 = 120a + 1926b \\ (-) \quad (-) \quad (-) \quad \underline{\hspace{1cm}} \\ -142 = \quad \quad -126b \end{array}$$

Therefore now we have $-142 = -126b$, this can be rewritten as $126b = 142$

Now, $b = 142/126 = 1.127$

Substituting the value of b in equation (iii), we get

$$\begin{array}{rcl} 160 & = & 8a + (120 * 1.127) \\ 160 & = & 8a + 135.24 \\ 8a & = & 160 - 135.24 \\ 8a & = & 24.76 \\ a & = & 24.76/8 \\ a & = & 3.095 \end{array}$$

Thus, we got the values of $a = 3.095$ and $b = 1.127$

Hence the required regression equation of Y on X:

$$Y = a + bX \Rightarrow \mathbf{Y = 3.095 + 1.127X}$$